



Optimizing LightGBM and XGBoost Algorithms for Estimating Compressive Strength in High-Performance Concrete

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Abstract

This study employed a regression approach to predict the compressive strength of high-performance concrete. We proposed the use of two machine learning algorithms: LightGBM and XGBoost. The models were enhanced by integrating them into the Optuna optimization framework and employing the tree-structured Parzen estimator as the optimization algorithm. The models were trained and validated on an original dataset. To reduce possible overfitting, the dataset was then augmented with SMOGN. K-fold cross-validation was also used to prevent overfitting. The performance of the models was evaluated using RMSE, MAE, and MAPE. A comprehensive analysis of the best-performing LightGBM and XGBoost models was performed based on the individual loss metrics using the SHAP technique to determine the influence of the individual mixture components on compressive strength. On the original dataset, XGBoost achieved an RMSE of 11.44, an MAE of 9.32, and a MAPE of 6.89, while LightGBM achieved an RMSE of 11.20, an MAE of 9.04 and a MAPE of 6.98. On the augmented dataset, XGBoost outperformed LightGBM on all metrics, with an RMSE of 5.67, an MAE of 2.83, and a MAPE of 2.06. LightGBM achieved an RMSE of 5.82, an MAE of 3.35, and a MAPE of 2.48. Both models identified micro steel fiber as the most important component in the original dataset, but disagreed on the second most important component, alternating between water and superplasticizer. However, with the augmented dataset, the SHAP analysis showed that both models consistently ranked micro steel fiber and water as the two most influential components.

Keywords Compressive strength · High-performance concrete · Machine learning · LightGBM · XGBoost · SHapley additive explanations

1 Introduction

Advances in materials and production technology have popularized special concretes with compressive strengths of about 200 MPa or more, referred to as high-performance concrete (HPC) [1–3]. HPC is more complex than high-strength

concrete (HSC). HSC primarily focuses on achieving high strength; HPC not only provides high strength but also offers lower permeability and more excellent durability, making it more challenging to design and manufacture [4, 5]. In this case, a well-designed mix is critical in ensuring that HPC performs as intended [6–9]. In particular, high cement content, low water–binder ratio, fine powders (quartz, silica fume, etc.), well-graded aggregates, and highly water-reducing admixtures are usually used to achieve high packing density since the packing density of concrete plays a significant role in determining its strength and durability [10]. Previous studies have highlighted the benefits of ternary blending, which involves using cement combined with two different supplementary cementitious materials (SCMs) to enhance these properties [11]. Incorporating SCMs into concrete increases the matrix's strength and density while simultaneously increasing its brittleness. At this stage, adding randomly dispersed fibers can effectively improve the flexural performance of thicker concrete structures [12, 13]. Steel

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fibers bonded with the matrix help restrain tensile stresses caused by pore water, reducing crack formation and improving durability. Additionally, steel fibers decrease concrete shrinkage, with fiber geometry playing a key role in influencing this behavior [14]. In short, unlike traditional concrete mixes, the design of the HPC mix includes many possible components and can be considered a multicriteria optimization process [15, 16]. However, optimizing particle size distribution to produce a densely packed microstructure is a major challenge in the design of HPC mixes [17, 18].

In addition to the challenges of developing HPC mixes, it is essential to note that the relationship between the engineering properties of HPC and their mix components is not strictly linear and is challenging to characterize using conventional statistical methods [19]. Indeed, researchers developed prediction models for the 28-day compressive strength of HPC using various statistical techniques, including linear regression (LR), multi-logistic regression (MLR), nonlinear regression (NLR), and response surface methodology (RSM) [20]. However, robust and advanced approaches are still being sought to explain the relevant experimental data and to model the relationship between mechanical properties and mix components [16, 19]. In particular, the development of novel methods to estimate the compressive strength of HPCs is becoming increasingly popular in research [21]. This is because laboratory testing of HPCs costs a lot of time and money to obtain a well-designed mix with the desired compressive strength [22]. In this regard, there have been many attempts to reduce the number of mix designs and experimental tests, including the use of empirical or computational modeling, statistical methods, and artificial intelligence (AI) approaches [1, 23–25].

AI-based machine learning (ML) approaches have recently experienced significant growth in compressive strength estimation due to their remarkable ability to identify nonlinear and ambiguous relationships between input (mix components) and output variables (compressive/tensile strength) in a dataset [26–28]. In this context, Han et al. 2019 claimed that a variety of ML algorithms, such as artificial neural networks (ANNs), support vector machine (SVM), decision trees (DT), and ensemble algorithms, have been used to estimate the compressive strength of HPC [29]. Lee et al. 2022 also mentioned that fully connected neural networks (FCNNs) are suitable for predicting the compressive strength of HPC because regression problems are the process of discovering relationships between inputs and outputs [21]. The research suggests that more sophisticated regression methods are needed for better results. Accordingly, regression-based models can be improved by ensemble learning methods composed of multiple base learners such as DT, neural networks (NN), or other supervised learning methods [30]. Overall, super learner models based on various ensemble methods, including Random Forest Regression (RFR),

adaptive boosting (AdaBoost), gradient boosting machine (GBM), extreme gradient boosting (XGBoost), light gradient boosting machine (LightGBM), and categorical gradient boosting (CatBoost) are attempted to solve these type of estimation problems [31]. The growing trend toward the use of combined strategies shows that ensemble approaches improve the applicability of individual AI techniques on the estimation of compressive strength [24, 32–35], as ensemble models perform better than individual ML models, as described in the literature [36].

In addition to selecting an appropriate model, the performance evaluation of the proposed models for estimating compressive strength is required. According to some previous studies, validation, and performance evaluation of the suggested models are generally accomplished using K-fold cross-validation, mean absolute error (MAE), mean square error (MSE), root mean square error (RMSE), coefficient of determination (R^2), relative root mean square error (RRMSE), performance index (PI), and Nash–Sutcliffe Efficiency (NSE), etc. [36, 37]. Generally, lower error values (MAE, MSE, RMSE, etc.) and higher R^2 values between the predicted and measured results indicate better model performance. Table 1 summarizes some of the recent compressive strength estimation studies conducted using this model validation approach.

All these studies agreed that the input variables significantly affect the outcome of the application of the model. Therefore, sensitivity analyses are performed to analyze the effects of input parameters [45] thoroughly. The sensitivity analysis is based on the contribution of the input variables to the prediction of the model results. However, the datasets used in such studies are usually from published articles. Therefore, it may be impossible to fully represent the performance of the proposed model due to limitations such as irrelevant input parameters, insufficient dataset, etc. This limitation has also been investigated, and the effect of input parameters on output parameters has been evaluated using ML model interpretation tools such as SHapley Additive exPlanations (SHAP) [27, 33, 45–47].

1.1 Aim and Scope

This study aims to employ both LightGBM and XGBoost ensemble models to predict the compressive strength of HPC. Unlike previous studies, this research is innovative in its use of optimization algorithms and data augmentation techniques to achieve the lowest mean absolute error (MAE), mean absolute percentage error (MAPE), and root mean squared error (RMSE), thereby improving model performance. In this study, using actual data also aims to compensate for limitations such as irrelevant inputs or insufficient data sets and ensure a more accurate model evaluation through tools such

Table 1 Previous studies on estimating compressive strength

Authors	Methods	Concrete type	Input	Dataset	Statistical metrics
Lee et al. 2022 [21]	Random Forest Regression (RFR), AdaBoost, Gradient Boosting Machine (GBM), XGBoost, LightGBM, CatBoost	High-Performance Concrete	For Dataset 1, Cement, Coarse Aggregate, Fine Aggregate, Water, Blast Furnace Slag, Fly Ash, Superplasticizer, Age of Testing. For Dataset 2, Fly Ash Replacement Ratio, Silica Fume Replacement Ratio, Total Cementitious Material, Fine Aggregate, Coarse Aggregate, Water, High Rate Water Reducing Agent, Age of Testing. For Dataset 3, Slump of Concrete, Water-to-Binder Ratio, Water, Fine Aggregate Ratio, Fly Ash Replacement Ratio, Air Entraining Agent, Superplasticizer. For Dataset 4, Cement, Coarse Aggregate, Fine Aggregate, Water, Superplasticizer	Dataset 1: 1030 Dataset 2: 144 Dataset 3: 104 Dataset 4: 357	MAE, MSE, RMSE, and R^2
Shen et al. 2022 [27]	XGBoost, AdaBoost, Bagging	Ultra-High-Strength Concrete	Cement, Fly Ash, Silica Fume and Silicate, Sand and Water, Superplasticizer, Steel Fiber, Steel Fiber Aspect Ratio, and Curing Time	372	MAE, RMSE, and R^2
Dai et al. 2022 [31]	Multiple-Layer Perceptron Neural Network (MLPNN), AdaBoost, Bagging,	High Strength Fiber-Reinforced Concrete	Cement, Fly Ash, Slag, Silica Fume, Nano-Silica, Limestone Powder, Sand, Coarse Aggregate, Maximum Aggregate Size, Water, Superplasticizer, Steel Fiber, Steel Fiber Diameter, Steel Fiber Length, and Curing Time	255	MAE, RMSE, and R^2



Table 1 (continued)

Authors	Methods	Concrete type	Input	Dataset	Statistical metrics
Shang et al. 2022 [38]	Decision Trees (DT), AdaBoost	Recycled Coarse Aggregate (RCA)-Based Concrete	Water, Cement, Fine Aggregate, Natural Coarse Aggregate, RCA, Superplasticizers, Water Absorption of RCA and Maximum Size of RCA, Density of RCA	344	MAE, MSE, RMSE and R^2
Yuan et al. 2022 [39]	Gradient Boosting and Random Forest Regression (RFR)	Recycled Aggregate Concrete (RCA)	Recycled Concrete Aggregate (RCA) Replacement Ratio, Parent Concrete Strength, Bulk Density of Natural Aggregate, Bulk Density of RCA, Bulk Density of Natural Aggregate, Water Absorption of Natural Aggregate, Water Absorption of RCA Aggregate–Cement Ratio (A/C), Effective Water Cement Ratio (W_{eff}/C), Nominal Maximum Natural Aggregate Size, Nominal Maximum RCA Size, Los Angeles Abrasion Index of Natural Aggregate, Los Angeles Abrasion Index of RCA	638	MAE, RMSE, and R^2
Nafees et al. 2021 [40]	Multiple-layer perceptron Neural Networks (MLPNN), Adaptive Neural Fuzzy Detection Systems (ANFIS), Genetic Programming	Silica Fume-Based Green Concrete	Cement, Fine Aggregate, Coarse Aggregate, Water, Superplasticizer, and Silica Fume	283	MAE, RMSE, RMSLE and R^2
Ahmad et al. 2021 [41]	Bagging, AdaBoost, Decision Trees (DT), Gene Expression Programming (GEP)	Concrete Containing Supplementary Cementitious Materials	Cement, Fly Ash, Blast Furnace Slag, Water, Superplasticizer, Coarse Aggregate, Fine Aggregate, and Age	1030	MAE, MSE, RMSE, R^2 , and K-fold cross-validation

as SHapley Additive ExPlanations (SHAP). Thus, the optimal model is further evaluated using SHAP values, which attribute contributions to each feature and help identify the key factors influencing compressive strength predictions. In summary, the combination of all strategies used in this study addresses the limitations of experimental work in developing

multicomponent mixtures, reducing human effort, time, and cost.



Table 1 (continued)

Authors	Methods	Concrete type	Input	Dataset	Statistical metrics
Song et al. 2021 [42]	Gene Expression Programming (GEP), Artificial Neural Network (ANN), Decision Trees (DT)	Concrete with Fly Ash Admixture	Cement, Fly Ash, Superplasticizer, Coarse Aggregate, Fine Aggregate, Water, and Age	98	RME, RMSE, R^2 , and K-fold cross-validation
Farooq et al. 2021 [43]	Artificial Neural Network (ANN), Decision Trees (DT), Random Forest Regression (RFR), Support Vector Machine (SVM), Multiple Linear Regressions (MLR), Bagging and Boosting	High-Performance Concrete	Cement, Water, Sand, Gravels, Superplasticizer, Concrete Age, Fly Ash, and Granulated Blast Furnace Slag	1030	MAE, RMSE, RMSLE, K-fold cross-validation
Xu et al. 2021 [44]	Random Forest Regression (RFR), AdaBoost, Support Vector Machine (SVM)	High-Performance Concrete	Fine Aggregate, Coarse Aggregate, Cement, Water, Superplasticizer, Fly Ash, and Blast Furnace Slag	425	MAE, RMSE, R^2 , and K-fold cross-validation

Table 2 The descriptive statistics of the dataset

Parameter	Minimum	Median	Maximum	Mean	Standard deviation
Cement (kg/m ³)	806	917	1285	955.068	138.239
Silica Fume (kg/m ³)	0	200	343	213.068	96.983
Quartz Sand (kg/m ³)	0	133	500	128.014	121.889
Superplasticizer (kg/m ³)	6	26	60	30.3562	15.912
Micro Steel Fiber (kg/m ³)	0	33	87	38.4384	30.707
Silica Sand (kg/m ³)	883	1167	1613	1180.01	219.564
Water (kg/m ³)	167	215	287	222.671	45.971
Compressive Strength (MPa)	103	132	175	132.137	14.725

2 Dataset

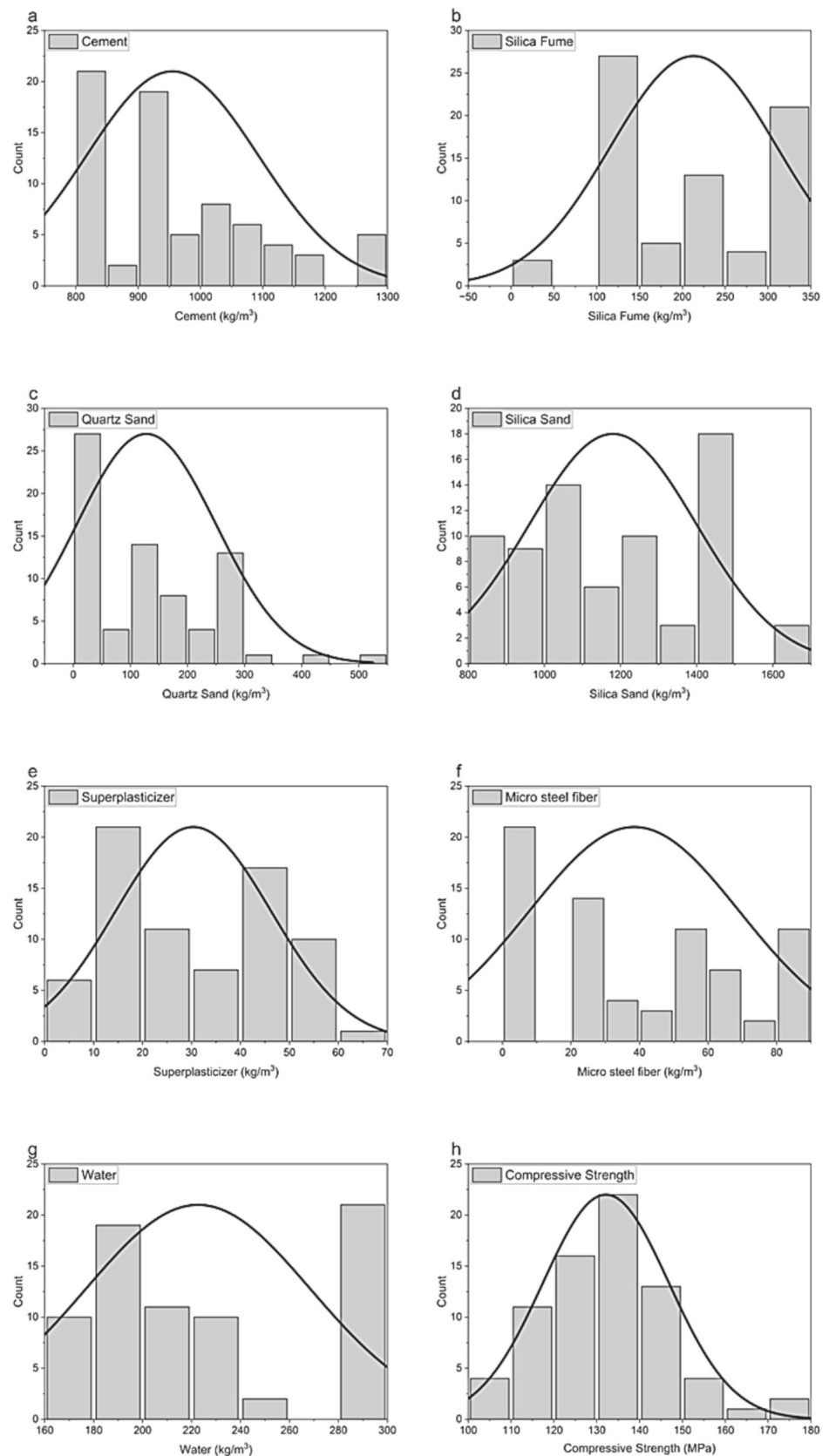
The dataset was obtained from an experimental study by the authors in this study. It included 73 different mixes with specific constituents, including CEM I 52.5 R white cement, silica fume, quartz sand, silica sand, new generation superplasticizer, micro steel fibers, and water. Then, the compressive strength of each sample ($10 \times 10 \times 10$ cm cube samples) was measured and defined as the outcome variable of each mixture. Table 2 contains descriptive statistics of the dataset, with seven input variables representing the mixture

of components and one output variable representing the compressive strength. Then, the relative frequency distribution of each input variable is shown in Fig. 1.

In addition, the correlation between all parameters was analyzed, and a graphical representation can be seen in Fig. 2 as a Spearman correlation heatmap. In Fig. 2, the correlation between input and output parameters is relatively low, generally ranging from -0.49 to 0.47. The range of correlation coefficients shows that the input variables can be considered low to moderately correlated with compressive strength. Therefore, the relationship between the proportional components such as water/cement, micro steel fiber content, superplasticizer/cement, water/silica fume and cement ratio,



Fig. 1 Relative frequency distribution of input variables: **a** cement, **b** silica fume, **c** quartz sand, **d** silica sand, **e** superplasticizer, **f** micro steel fiber, **g** water, and **h** compressive strength



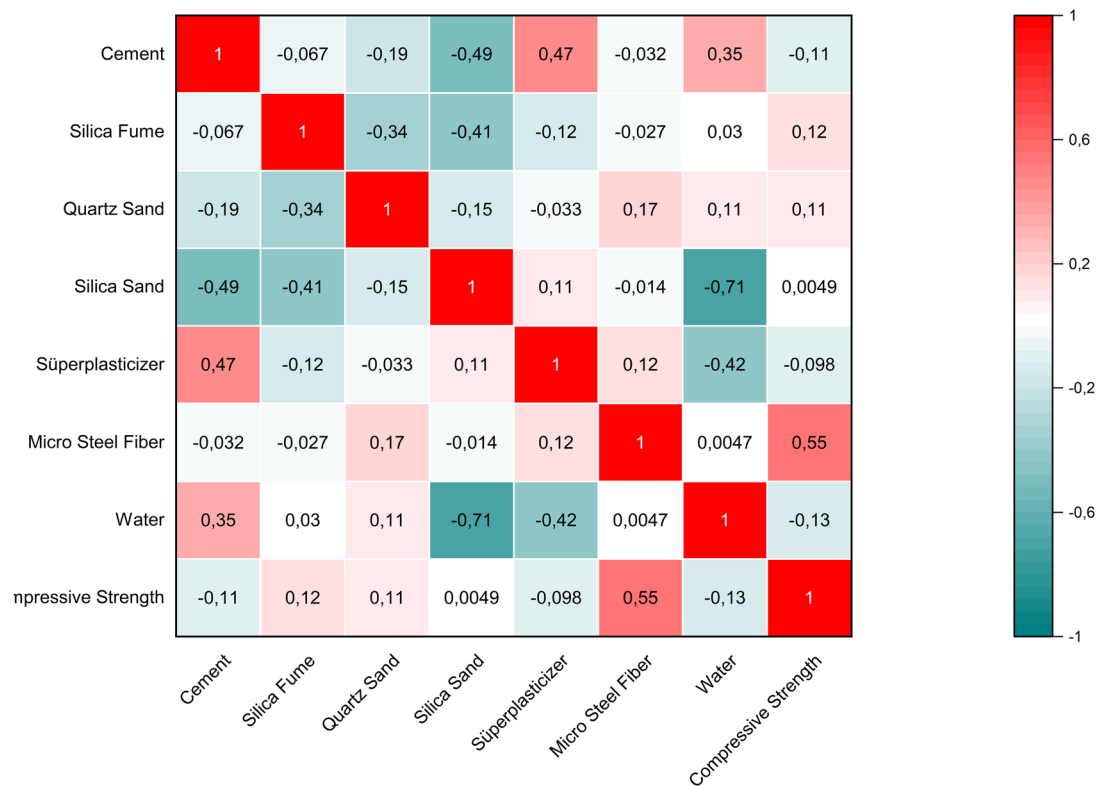


Fig. 2 Spearman's correlation heatmap

and compressive strength of the mixtures is also summarized in Fig. 3 to better characterize the dataset. The figures also include a graphical representation of the regression results using confidence and prediction ranges.

3 Methodology

This section provides a concise overview of this study's methodology and experimental design. The problem to be studied is the determination of the compressive strength value, which is a regression problem in the field of ML. To address this problem, we have selected two of the most robust and powerful regression algorithms, LightGBM [48] and XGBoost [49], as our ML models. Both algorithms rely on pre-defined default parameter values for their initial configuration. Manually tuning parameters is time-consuming. Therefore, it is widely recognized that parameter optimization is a crucial consideration in ML research [50]. We have conducted thorough parameter optimization to ensure the best possible performance of the LightGBM and XGBoost models for our study. For this purpose, the Optuna framework was utilized [51]. The Optuna framework is an open-source framework tailored to optimize ML models' hyperparameters [52–54]. It is an advanced tool that utilizes state-of-the-art algorithms to explore the extensive hyperparameter

space of complex models efficiently. Moreover, we selected SHAP to fully explain the behavior of the best LightGBM and XGBoost models optimized by Optuna framework. In the context of ML, the SHAP value has been adapted to explain individual features' contribution to a predictive model's output. It measures the average marginal contribution of each feature to the final prediction over all possible subsets of features. The following subsections give brief information about the methods employed in this study.

3.1 LightGBM (Light Gradient Boosting Machine)

LightGBM algorithm is a gradient-boosting decision tree algorithm (GBDT) proposed in 2017. It is a decision tree model and has been used for different kinds of ML problems, such as classification and regression. The main advantage of the LightGBM algorithm is to speed up the task at hand [48]. While the LightGBM algorithm is fundamentally based on the GBDT algorithm, it introduces several vital enhancements that improve the original approach. One of the primary challenges of GBDT is the time required to construct trees, mainly when working with large datasets. LightGBM addresses this issue through a histogram algorithm, enabling efficient binning of continuous features and resulting in significant time savings during the tree-building process [55]. Another significant benefit of the histogram



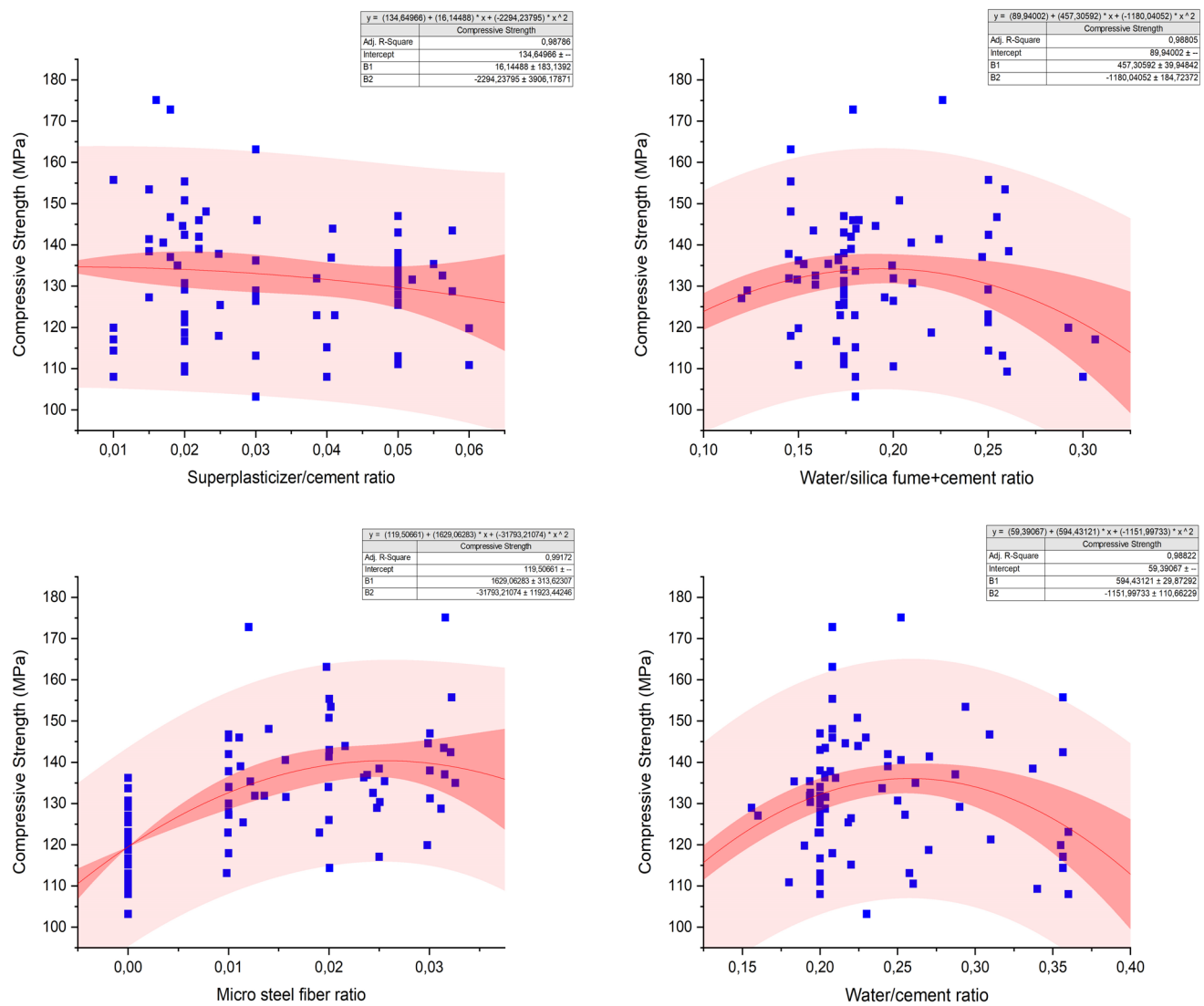


Fig. 3 The relationship between the proportional components and the compressive strength of the mixtures

algorithm employed by LightGBM is its ability to provide regularization effects. Specifically, the histogram algorithm reduces the variance of DT, which can help to prevent overfitting. This is accomplished by aggregating similar data points into a single bin, reducing the DT's granularity and reducing its tendency to overfit small-scale, idiosyncratic patterns in the data [56].

3.2 XGBoost (Extreme Gradient Boosting)

The XGBoost algorithm uses a supervised learning paradigm in which gradient boosting is used to construct an ensemble of a number of decision trees [49]. During training, the algorithm starts with a first prediction and then computes the residuals, representing the difference between the prediction and the observation. It significantly improves conventional

gradient-boosting models by incorporating second-order gradient information, enhancing the optimization process. Each tree sequentially adapts its predictions in the algorithm to minimize the residuals between the previous tree's output and the true values. This iterative process continues until a specified stopping criterion is met. Ultimately, the final predictions are obtained by summing the results from all the individual trees.

3.3 SMOGN (Synthetic Minority Over-Sampling Technique for Regression with Gaussian Noise)

Although ML algorithms often demonstrate enhanced performance when trained on substantial datasets, real-world datasets frequently encounter limitations in terms of data quantity. This can markedly influence the performance of ML algorithms. Moreover, acquiring additional data in real-world

scenarios can be time-consuming and resource-intensive, if not impractical. In such instances, data augmentation techniques emerge as a viable strategy for researchers. Data augmentation techniques have been extensively studied in the context of binary classification problems, with reliable algorithms such as SMOTE (Synthetic Minority Over-sampling Technique) readily available [57]. Since the problem at hand involves regression, applying SMOTE is not appropriate. Instead, we utilize SMOGN (Synthetic Minority Over-Sampling Technique for Regression with Gaussian Noise) [58] to augment the dataset, ensuring more robust and balanced training data for our study. The approach integrates random under-sampling with two over-sampling techniques: SmoteR and the incorporation of Gaussian noise. SMOGN combines two different sampling techniques, thus enabling data diversity. It combines one random under-sampling and two over-sampling techniques. New data points are generated either through K-nearest neighbors (KNN) or by introducing Gaussian noise. To the best of the authors' knowledge, this method is being applied to a dataset in this field for the first time.

3.4 Model Optimization

Every ML algorithm is characterized by specific hyperparameters, which may have varying ranges of acceptable values. To construct a solid ML system effectively, these hyperparameters must be optimized to obtain the best possible performance [56]. In many cases, default hyperparameter values can lead to suboptimal results, as these values may not be appropriate for a given dataset or problem. When it comes to optimizing an ML model, there are generally two broad categories of techniques that can be employed. The first of these categories can be referred to as a “trial and error” approach, in which the values of hyperparameters are selected through manual experimentation and evaluation. However, the manual trial and error approach to hyperparameter optimization can be impractical in many cases due to the large and often complex hyperparameter spaces typical of ML models. The range of possible values for each hyperparameter can be broad and highly variable, and manually searching this space can be time-consuming, inefficient, and prone to human error. Another approach involves the automatic hyperparameter optimization of an ML model, thereby dispensing with the manual approach. Automated optimization techniques are often advantageous in speed and may increase the likelihood of discovering an optimized model more efficiently.

The Optuna framework offers automated hyperparameter optimization for ML models. Due to its tree-based nature, manually designing the LightGBM and XGBoost algorithms' numerous parameters can be time-consuming. For

this reason, the Optuna framework was selected as a hyperparameter optimization framework for both models. In an automated hyperparameter optimization approach, selecting the next set of hyperparameters poses a challenge. The automated approach can either opt for blind selection, a random search, or a specific mechanism to yield a more coherent set of hyperparameters. Given the promising nature of employing a specific mechanism, the tree-structured parzen estimator (TPE) was chosen as the estimator algorithm for selecting the next set of hyperparameters [59]. Unlike the random search, which lacks prior information regarding its choices of hyperparameter sets, TPE utilizes Bayesian inference and incorporates previous selections to partition the hyperparameter space into “bad” and “good” sections. As the algorithm progresses, it strategically explores the “good” section of the hyperparameter space, ultimately discovering the optimal hyperparameters. Figure 4 presents a graphical comparison between random sampling and sampling with the TPE method.

3.5 Model Evaluation

Every ML model relies on a cost function that governs its performance and guides the optimization process. Since the problem is to predict a continuous value, it can be categorized as a regression problem. Thus, suitable cost functions for the problem were utilized [60]. The classical root mean squared error (RMSE) cost function was selected to specify the error rate between the actual compressive strength value and the predictions of both models. Moreover, to fully assess the performance of models employed in our study, we also selected mean absolute error (MAE) and mean absolute precision error (MAPE) metrics. The formulas for the RMSE, MAE, and MAPE are given in Eq. (1), Eq. (2), and Eq. (3), respectively.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (1)$$

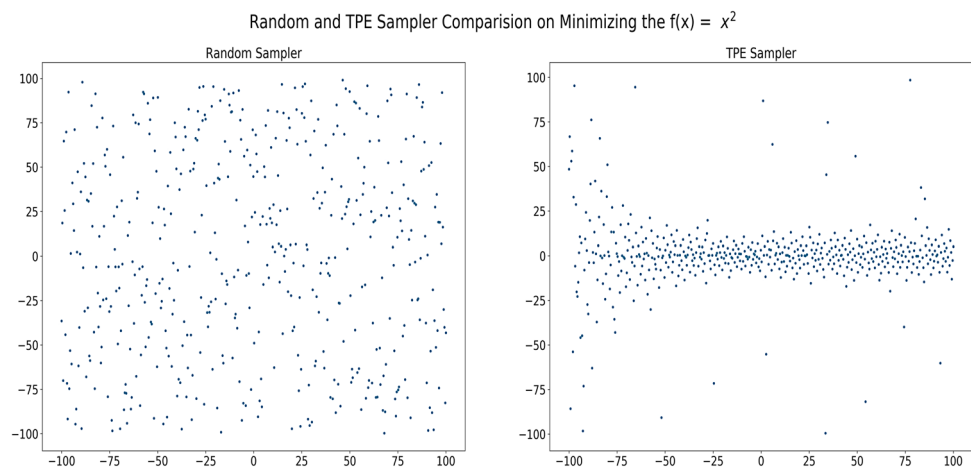
$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (2)$$

$$\text{MAPE} = \frac{100\%}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (3)$$

where $\hat{y}_i$ is the predicted value of the ML model, y_i is the actual compressive strength, N is the total number of points in the dataset. Overall, lower values for each metric indicate better model performance.



Fig. 4 A comparison between random sampler and TPE sampler



3.6 Model Interpretation

In the literature on ML, there is an increasing emphasis on model interpretation [61–63]. It is not enough for researchers to just display the model with the lowest error rate. Explanations that elucidate the underlying mechanisms, insights, and decision-making processes of the model are required. This enables a deeper understanding of its inner workings and enhances the transparency and trustworthiness of the results. Therefore, it is not enough to suggest an ML model alone; one must also analyze it to gain a more profound understanding and maximize its potential. For this reason, the SHAP value is proposed to interpret the LightGBM and XGBoost algorithms. Within the domain of ML, SHAP value serves as a tool for quantifying the significance and contribution of individual features or variables within a predictive model. It offers a systematic approach for attributing recognition to each feature by evaluating its influence on the model's predictions.

3.7 K-Fold Cross-Validation

A classical approach in the modeling phase of ML involves partitioning the data into two distinct parts: one for training the model and the other for evaluating the model's performance. However, maintaining a fixed split between training and evaluation data can lead to suboptimal model generalization, especially when working with limited data. To enhance generalization, an ML model is recommended to utilize the entire dataset across both training and testing phases. This ensures that all data is processed, which is achieved through cross-validation techniques. This approach involves partitioning the dataset into multiple folds or subsets known as K. The model is then trained on one portion, the training set, and evaluated on the remaining data, referred to as the validation or test set. Figure 5 gives a figurative explanation of the K-fold cross-validation employed in the study.

Table 3 Parameters of the LightGBM and XGBoost for the optimization process

Parameter name	Lower bound of search space	Upper bound of search space
Number of estimator	50	300
Learning rate (LR)	0.01	0.05
Max depth	2	300
Drop rate	0.0	0.1
Reg_Lambda	0	2

By repeating this process multiple times with different subsets, cross-validation provides a more comprehensive assessment of the model's performance, reducing reliance on a single train-test split.

3.8 Experimental Settings

For the experiments, the Python programming language (3.6.10) was used. LightGBM, XGBoost, Optuna, and SHAP libraries were utilized for the ML models, optimization framework, and model interpretation, respectively. The overall experimental framework is divided into two phases. First, the original (non-augmented) dataset was evaluated using both LightGBM and XGBoost. A thorough optimization process was conducted, with each model being individually optimized for the respective loss functions mentioned above. The same procedure was repeated in the second phase using the augmented dataset. Given the substantial number of parameters in both the LightGBM and XGBoost algorithms that must be configured prior to training, a specific set of core and shared parameters was selected for optimization using the Optuna framework. The parameters selected for optimization and their respective search spaces are given in Table 3.

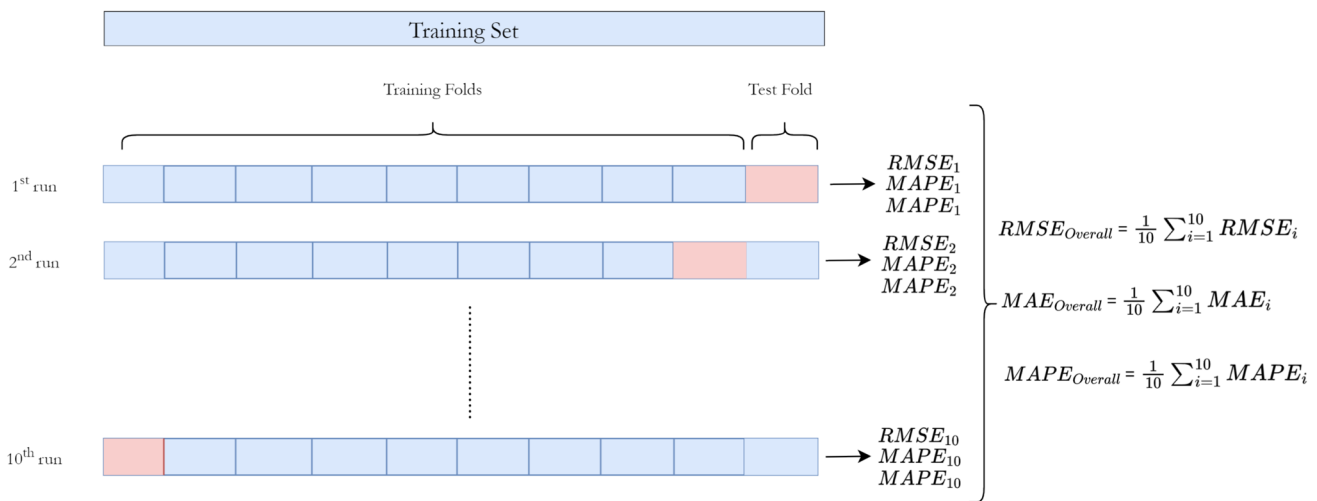


Fig. 5 K-Fold cross-validation

During the parameter selection process, a brief explanation can be provided for the Drop Rate and regularization term (Reg_Lambda) parameters. Given the small size of the experimental dataset, both models are prone to overfitting, despite the use of K-fold cross-validation. These parameters help control the models and prevent potential overfitting. Therefore, we allowed Optuna to optimize these two parameters to reduce the likelihood of overfitting. Briefly, regularization term (Reg_Lambda) applies L2 Regularization to models while Drop Rate controls the fraction of trees to be dropped. Since the current task involves a regression problem, the regression versions of both LightGBM and XGBoost algorithms were employed. Regarding data preprocessing, the variable chosen to be predicted was the compressive strength. The rest of the variables were considered as independent variables. An additional preprocessing step involved normalizing the independent variables, as it is widely recognized that normalization accelerates the convergence of ML models. K-Fold Cross-Validation technique was employed to assess the general performance of the model. The value of K was set to 10, indicating that the entire dataset was partitioned into 10 folds, with one-fold being held out as the validation set in each iteration. The mean of the RMSE values was computed across the 10 Folds, serving as an aggregate score for the model using the parameters selected through the Optuna framework. The same process was repeated for the MAE and MAPE values. Consequently, both ML models were optimized using different loss functions.

The Optuna framework was configured with 1000 trials, indicating that the framework executed 1000 iterations, with each iteration employing distinct parameters for the LightGBM and XGBoost models. The TPE sampler was chosen for the Optuna framework, leveraging a Bayesian approach

for parameter selection. Throughout the trials, the best configuration for each ML model—determined by the lowest RMSE, MAE, and MAPE values—was retained for further analysis and interpretation. Subsequently, the selected best models for each loss function were reevaluated and interpreted using SHAP values. The overall experimental schema is given in Fig. 6.

4 Results and Discussion

The ML models developed in this study were assessed using the MAE, MAPE, and RMSE metrics. As previously noted, the first fold of the experiments utilized the original dataset, comprising 73 data points. Table 4 presents the optimal results achieved when the models were individually optimized for each of these loss metrics during the initial stage of the experiments.

As illustrated in Table 4, each model demonstrated a relatively consistent convergence point when optimized for different loss metrics. However, the optimized parameters for each model exhibited substantial variation. XGBoost converged to the optimal point for each loss metric using a shallower tree depth than LightGBM. However, XGBoost required a more significant number of estimators than LightGBM. Despite minor variations in the loss metrics, each model exhibited comparable performance. LightGBM achieved RMSE, MAE, and MAPE values of 11.20, 9.04, and 6.98, respectively. Similarly, XGBoost yielded RMSE, MAE, and MAPE values of 11.44, 9.32, and 6.89, respectively. Results given in Table 4 solely show the optimal results achieved by the models. It lacks showing the effect of those parameters during optimization. To analyze the impact of each parameter during the optimization, we presented the



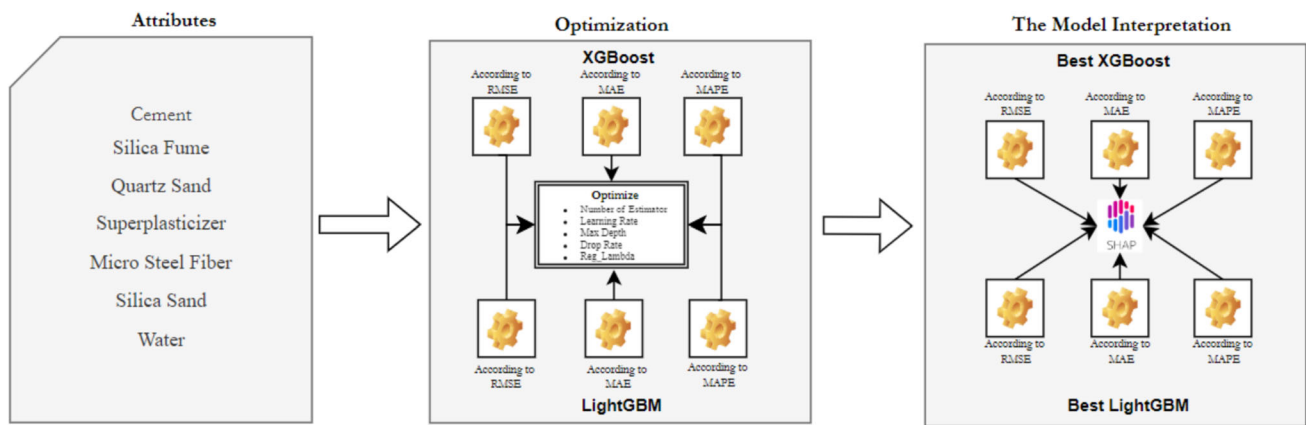


Fig. 6 Overall experimental schema

Table 4 Results of the optimization on the original dataset

		Optimized for RMSE	Optimized for MAE	Optimized for MAPE
Learning rate	XGBoost	0.045	0.037	0.041
	LightGBM	0.041	0.044	0.048
Drop rate	XGBoost	0.608	0.05	0.61
	LightGBM	0.0007	0.02	0.013
Number of estimator	XGBoost	118	116	164
	LightGBM	108	54	56
Req_Lambda	XGBoost	0.55	1.20	0.95
	LightGBM	0.2	0.15	0.70
Max depth	XGBoost	2	57	2
	LightGBM	191	177	271
Optimized loss	XGBoost	11.44	9.32	6.89
	LightGBM	11.20	9.04	6.98

parameter importances for each model when optimized for difference loss functions in Fig. 7.

As illustrated in Fig. 7, a more in-depth analysis of the models' optimization behavior revealed that, due to the relatively small size of the original dataset used in the first stage of the experiment, LightGBM exhibited a more substantial reliance on the regularization parameters Reg_Lambda and Drop Rate. Conversely, XGBoost demonstrated a greater emphasis on the number of estimators and learning rate. This apparent disparity can be attributed to the underlying mechanisms of these ML models. Furthermore, feature importance for both models indicated no significant differences across the various loss metrics. To gain deeper insights, SHAP interpretation was employed to analyze the models' behavior comprehensively. The most effective models for each metric were selected, followed by an evaluation of feature importance for the dataset, feature correlation, and the individual impact of each feature based on SHAP values. The results of this analysis are presented in Figs. 8, 9, and 10, respectively.

As illustrated in Fig. 8, each model prioritized micro steel fiber and superplasticizer for the original dataset. These attributes can be considered the most influential attributes based on the experimental data from this study. However, the models exhibited varying perspectives regarding the third most influential variable. According to Fig. 8, the LightGBM weighed more importance on Water, while XGBoost emphasized Quartz Sand. Both models exhibited consensus in identifying Silica Sand, Silica Fume, and Cement as less influential features for the task. Another noteworthy observation is that each model ranked features differently depending on the loss metric. This issue may be considered a significant factor in the selection of models and error functions.

Figure 9 shows the attribute's effect on the output: Compressive Strength. Both models consistently agreed that reducing the quantity of micro steel fiber leads to decreased compressive strength. A positive correlation existed between quartz sand and compressive strength. Both models emphasized that an increase in the strength of the superplasticizer

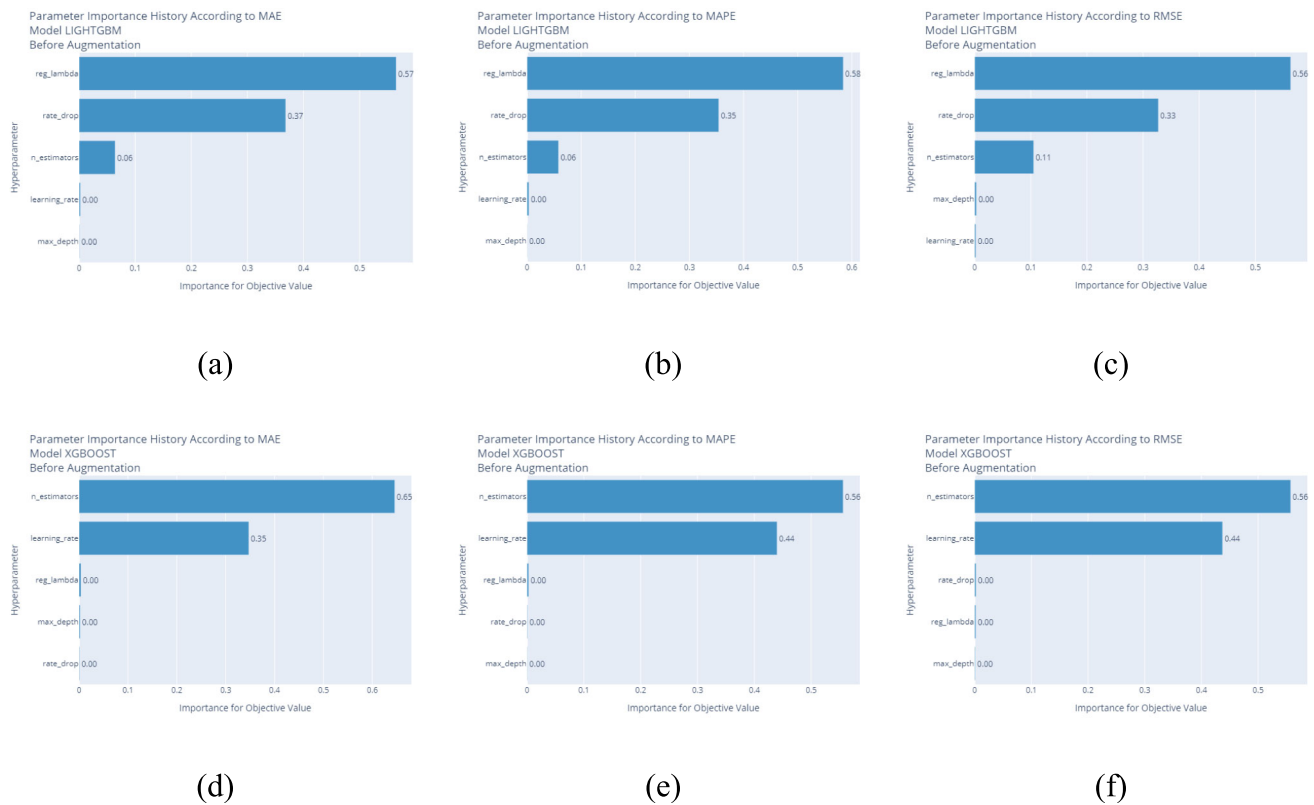


Fig. 7 Parameter importance of the models on the original dataset **a** LightGBM-MAE **b** LightGBM-MAPE **c** LightGBM-RMSE **d** XGBoost-MAE **e** XGBoost-MAPE **f** XGBoost-RMSE

is correlated with a decrease in compressive strength. Given LightGBM’s prioritization of water over quartz sand, Water may negatively impact compressive strength. However, for XGBoost, Water had a mixed influence on the output when the MAPE and RMSE metrics were used for optimization. This ambiguity can likely be attributed to the relatively small size of the dataset. XGBoost may have struggled to discern the effect of water accurately. Features such as cement, silica sand, and silica fume did not appear to have a significant or reliable impact on compressive strength, as evidenced by both models.

To conclude the analysis of the first stage of the experiment, Fig. 10 illustrates the feature correlations as determined by SHAP Values. For LightGBM, SHAP analysis revealed a stronger correlation between Water and Superplasticizer than other feature combinations. Interestingly, LightGBM exhibited a relatively consistent feature emphasis across all metrics. In contrast, XGBoost focused on different features depending on the specific loss metric. The RMSE-based feature correlation matrix for XGBoost showed that the water–silica sand correlation is higher than the water–superplasticizer, unlike the MAE and MAPE matrices. This indicates that RMSE emphasized the water–silica sand interaction more when reducing prediction errors, while MAE and MAPE focused more on the water–superplasticizer.

In the second stage of the experiments, SMOGN was employed to augment the original dataset. It is crucial to ensure that the augmented data maintains a distribution that closely resembles the original dataset. Accordingly, a comparison of the original and augmented datasets is presented in Fig. 11.

Following data augmentation, the augmented dataset comprised 247 samples. As illustrated in Fig. 11, the attributes of the augmented dataset (denoted as “Modified” in Fig. 11) exhibited a distribution that closely resembled the original dataset’s attributes. Fig. 11 demonstrates that the augmented data accurately represents the original distribution, thereby enabling their reliable use in assessing the performance of the ML models. Consistent with the procedures applied to the original dataset, the same methodology was employed for the augmented dataset. Following the same presentation schema as the initial approach, Table 5 presents the optimization results obtained from the augmented dataset.

A significant initial observation is that all loss metrics notably decreased as the number of samples increased. When the original dataset was employed, the error metrics exhibited a narrower range, with LightGBM demonstrating superior performance compared to XGBoost. However, as the number of samples increased, XGBoost outperformed LightGBM

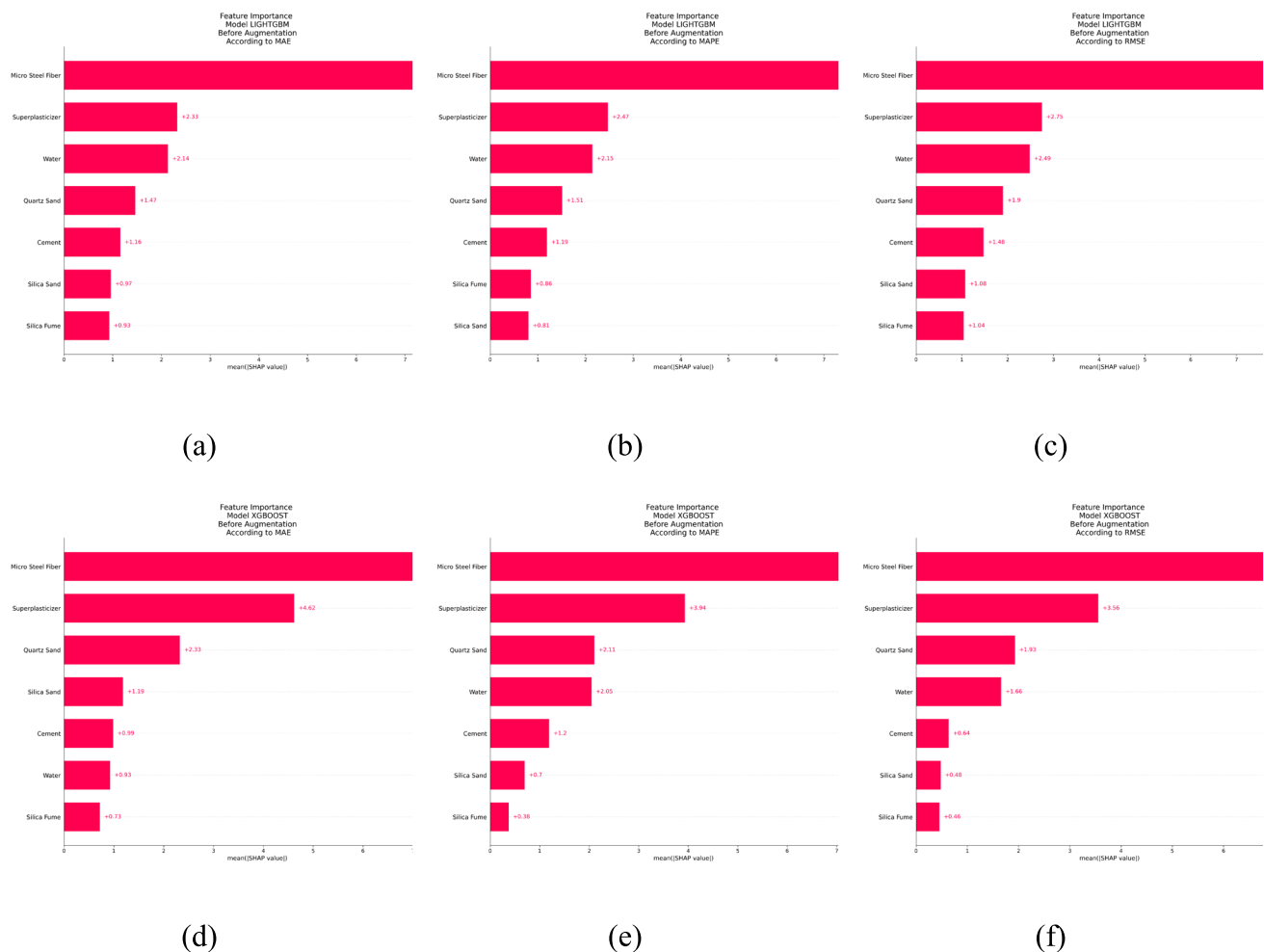


Fig. 8 Parameter importance of the models according to SHAP values on the original dataset **a** LightGBM-MAE **b** LightGBM-MAPE **c** LightGBM-RMSE **d** XGBoost-MAE **e** XGBoost-MAPE **f** XGBoost-RMSE

across all loss metrics. The Drop Rate parameter for LightGBM exhibited a decrease when the augmented dataset was used, while the same parameter for XGBoost did not show significant differences. Figure 12 presents the parameter importance of the models according to the respective loss metrics to elucidate the differences between optimization processes when using augmented data.

As illustrated in Fig. 12, despite a reduced Drop Rate for the LightGBM model when trained on the augmented data, its significance as a parameter remained substantial. Furthermore, the parameter importance remained relatively unchanged, even with the increased number of samples. Based on these findings for LightGBM, it can be concluded that the model effectively mitigated overfitting in both scenarios. Similarly, the parameter importance for XGBoost also remained nearly constant. When Figs. 7 and 12 were analyzed together, even though both models' error rates for each metric were close, it should be noted that both models converged at the same point, optimizing different parameters.

Figure 13 shows the feature importance for the ML models according to SHAP Values when trained on the augmented dataset. Compared to Fig. 8, both models agreed on feature importance except for minor deviations. Both models revealed micro steel fiber as the most important feature, followed by water. Interestingly, the choice of loss metric continued to influence the relative importance of features in LightGBM, with similar trends observed in XGBoost.

In Fig. 14, The SHAP diagrams for both LightGBM and XGBoost models following augmentation (using different error metrics) demonstrate how different features influence model outputs. It presents a more conventional assessment of attribute importance compared to Fig. 9. Consistent with previous findings, a reduction in micro steel fiber mainly correlates with a decrease in compressive strength. However, before augmentation, micro steel fiber mostly demonstrated an extensive and inconsistent influence in both positive and negative directions. After augmentation, its effect becomes

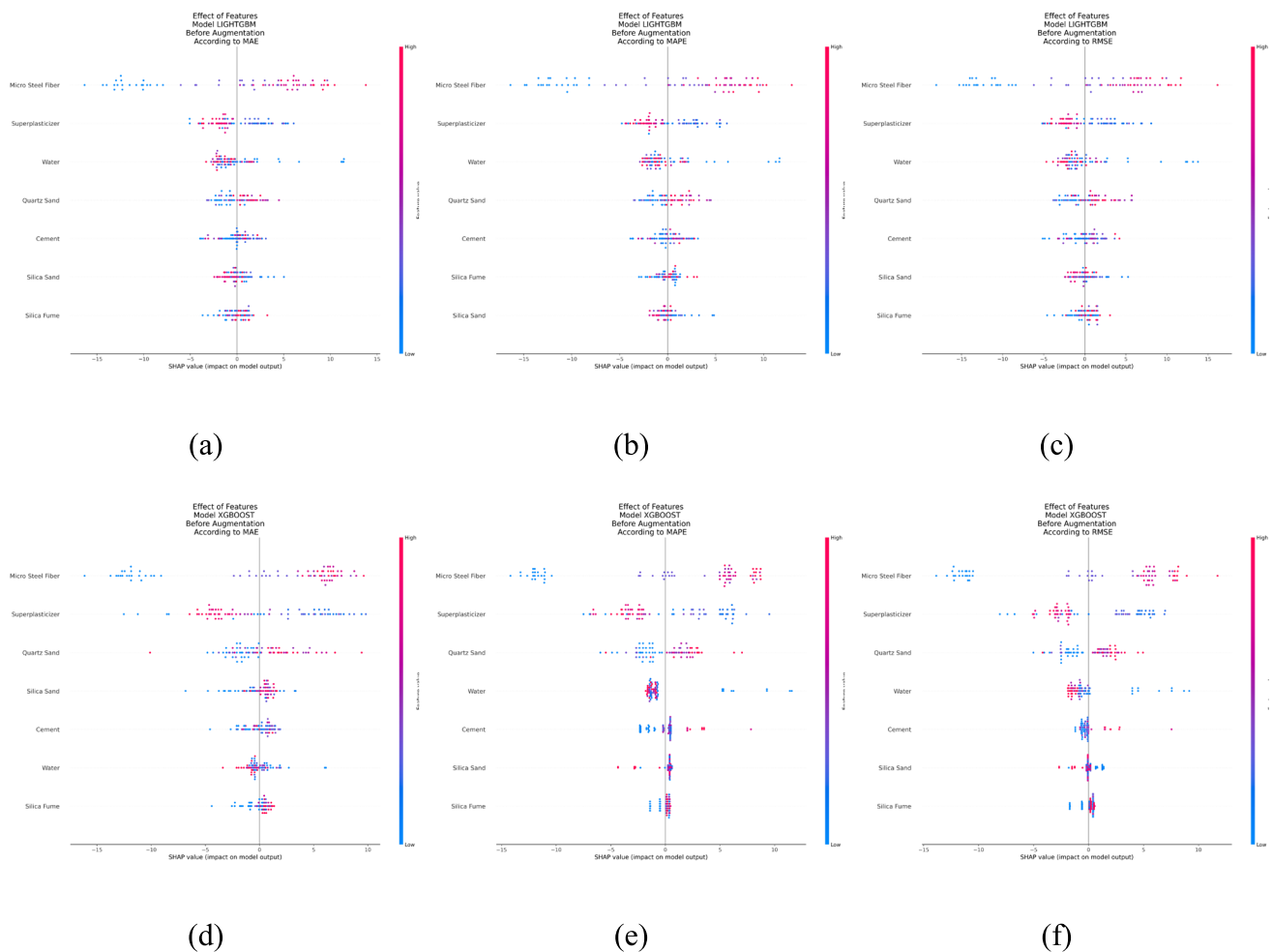


Fig. 9 Attributes effect on the output according to SHAP values on the original dataset **a** LightGBM-MAE **b** LightGBM-MAPE **c** LightGBM-RMSE **d** XGBoost-MAE **e** XGBoost-MAPE **f** XGBoost-RMSE

more focused and consistently positive, with reduced variability, ultimately enhancing the model's reliability. After augmentation, the more concentrated and consistent effects are observed not just for micro steel fiber but across all mix attributes, reflecting a general improvement in the model's reliability and performance. For example, before augmentation, the SHAP values for water are relatively balanced around zero, so its impact on predictions is unclear. In contrast to previous findings, after augmentation, the water attribute impacts compressive strength significantly. As illustrated in Fig. 14, reducing water use correlates with increased compressive strength. In conclusion, while micro steel fiber and water are the most essential features, other attributes such as superplasticizer, cement, and quartz sand significantly impact the model predictions. Interestingly, silica fume and silica sand have relatively minor effects across all models and metrics. To finalize the analysis of both models on the augmented data, feature correlation matrices are also provided in Fig. 15 for each loss metric.

Although minor deviations exist between models and loss metrics, each model constantly correlates water and micro steel fiber, as shown in Fig. 15. In the LightGBM model, water and superplasticizer dominate the feature interactions. In contrast, in the XGBoost model, micro steel fiber and water mostly showed stronger correlations. This distinction highlights how the models prioritize different feature interactions after the augmentation. At the end of the study, the comprehensive experimental schema revealed that XGBoost outperformed LightGBM in predicting compressive strength based on the employed loss metrics. Furthermore, our study demonstrated that the limited size of the dataset significantly influenced the outcomes, not only in terms of loss metrics but also in terms of potential inaccuracies in correlation and feature importance assessments.



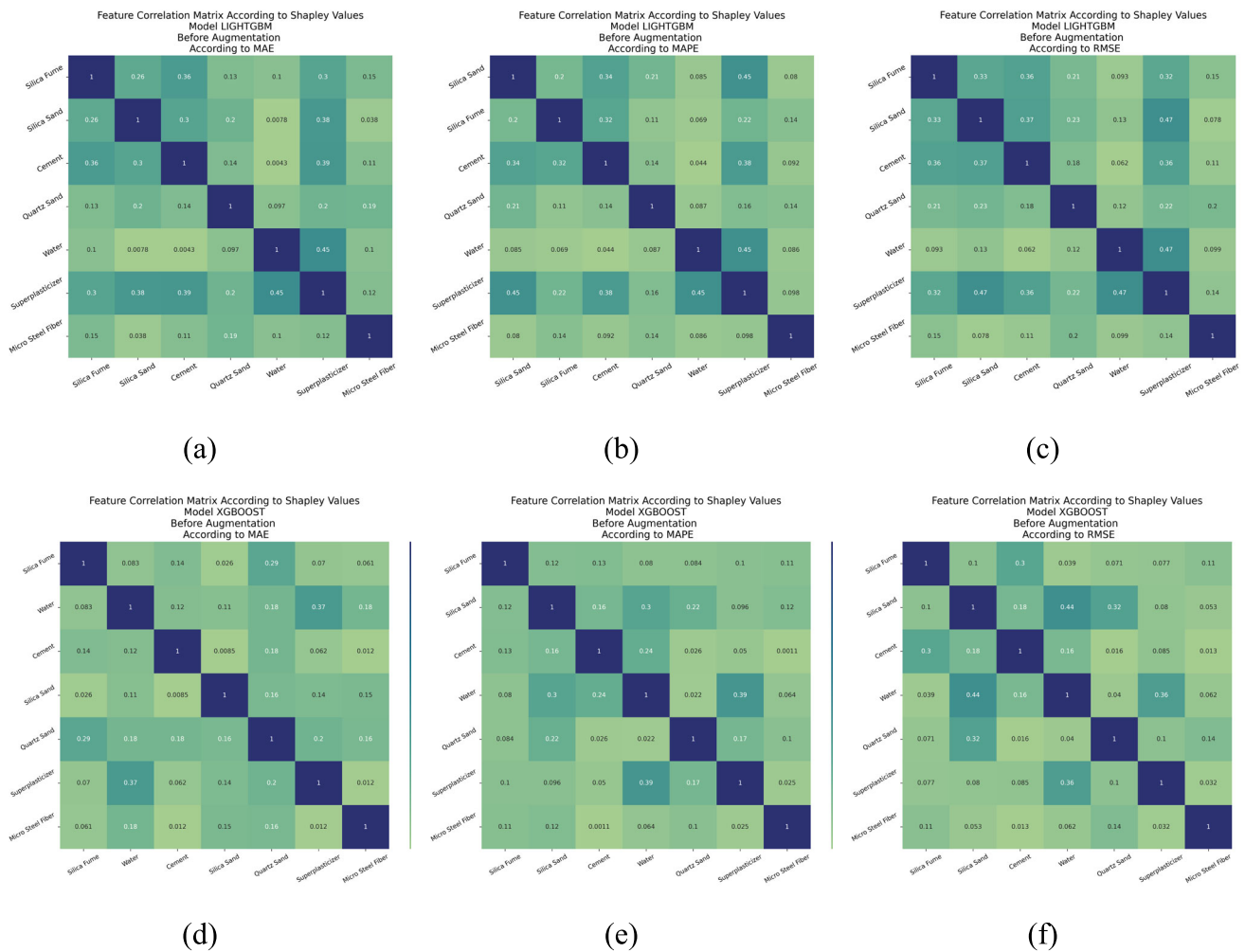


Fig. 10 Feature correlation matrix according to SHAP values on the original dataset **a** LightGBM-MAE **b** LightGBM-MAPE **c** LightGBM-RMSE **d** XGBoost-MAE **e** XGBoost-MAPE **f** XGBoost-RMSE

5 Practical Implications for Concrete Mix Design Optimization

The findings of this study have important implications for optimizing concrete mix design in HPC. By using optimized LightGBM and XGBoost models to predict compressive strength accurately, practitioners can streamline the production of HPC mixes that meet strength requirements and reduce the need for extensive physical testing. This reduction in testing can save significant time and resources, improving efficiency in construction project planning. Additionally, the SHAP analysis enhances the practical applicability of the models by identifying the most influential mix components affecting compressive strength, such as micro steel fibers and superplasticizers. These insights enable engineers to effectively adjust these critical parameters in mix designs, tailoring concrete properties to meet specific project requirements. For example, increasing the content of micro steel fibers by

10 kg/m³ can result in an average improvement in compressive strength of approximately 2.5 MPa. Similarly, variations in superplasticizer content can show a significant impact, with a 5 kg/m³ increase potentially leading to a change in compressive strength of up to 1.8 MPa. These numerical findings empower engineers to refine mix compositions by focusing on the most influential components, thereby optimizing material usage without compromising strength and durability. In summary, integrating this ML-based approach into the mixture design process offers multiple advantages. It enables faster decision-making by providing immediate feedback on mix designs, reduces variability from human error, and promotes sustainability by focusing on critical components, minimizing material waste and environmental impact.

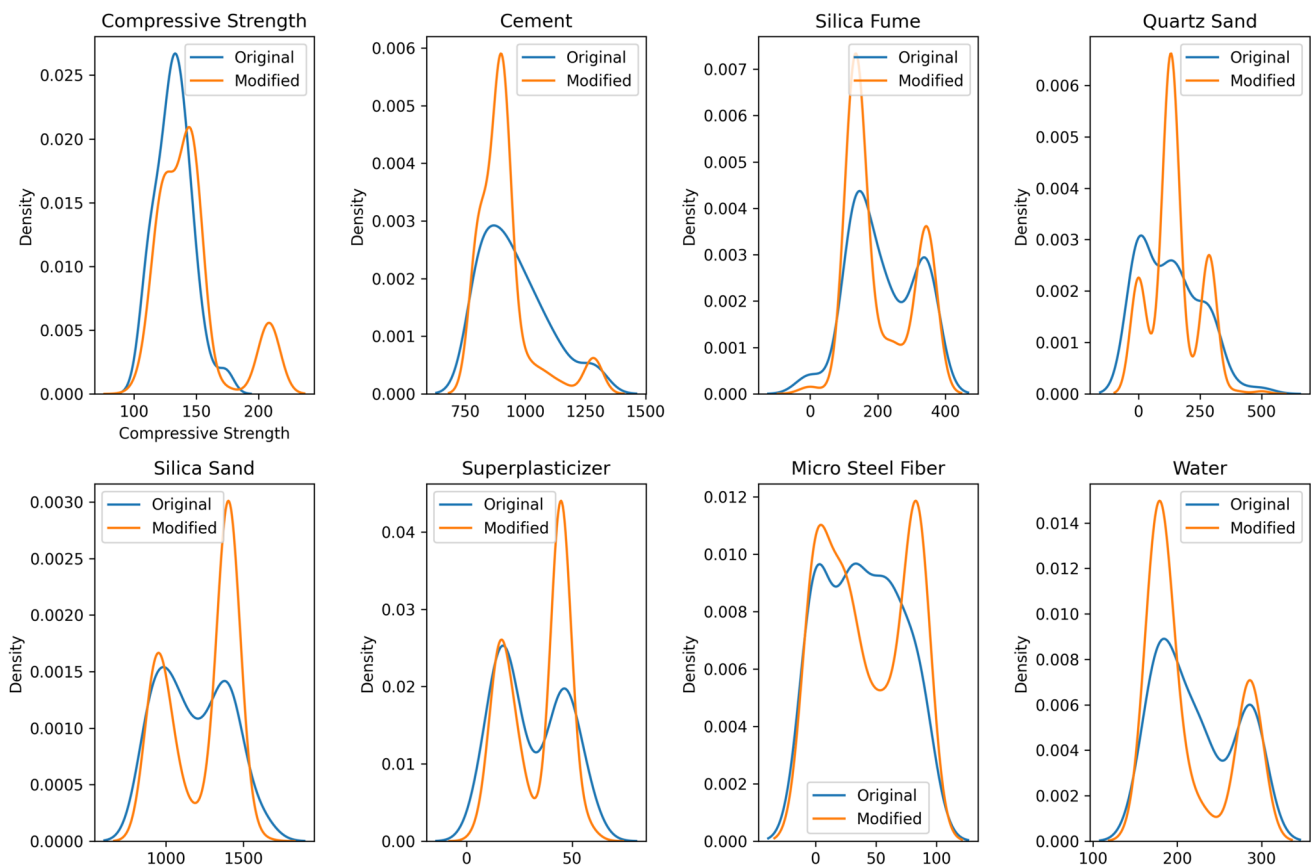


Fig. 11 Distribution of the variables of both the original dataset and the augmented dataset

6 Conclusion

This study illustrates the successful use of ML models, notably LightGBM and XGBoost, to estimate the compressive strength of HPC. Combining the Optuna framework for hyperparameter optimization with SMOGN for data augmentation achieved significant gains in prediction accuracy. Based on the study's results, the above input parameters, including cement, silica fume, quartz sand, silica sand, superplasticizer, micro steel fibers, and water, can be identified as important factors for predicting the compressive strength of HPCs. Accordingly, the study also provides the following specific results:

- It can be concluded that the automatic optimization of hyperparameters is crucial to achieving optimal performance in ML models. In this context, the Optuna framework, used as an automated optimization framework for the LightGBM and XGBoost algorithms, provided an efficient approach by employing the tree-structured parzen estimator (TPE) algorithm, which strategically explores the “good” part of the hyperparameter space to discover the optimal hyperparameters and improve the overall performance of the model. Additionally, the importance of

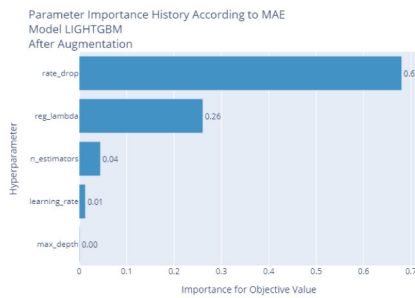
hyperparameters was evaluated to provide a comprehensive understanding of the models and their behavior during optimization. It is demonstrated that, although both ML models are similar, they converge at different points, optimizing distinct hyperparameters.

- The study evaluated the feature correlation matrix, attribute effects on output, and feature importance based on SHAP values for the ML models before and after data augmentation. These comprehensive evaluations enhance the reliability of the models, providing a deeper understanding of how mix components influence the compressive strength parameter. The study further demonstrated that correlations between features may be misinterpreted when working with a limited sample size and should be interpreted cautiously.
- Based on the feature importance charts for the LightGBM and XGBoost models across all error metrics, micro steel fiber, and water become more dominant after data augmentation. This is evidenced by the increased mean SHAP values for these features, indicating their growing influence on the model's predictions after augmentation. The impact of data augmentation was reflected in a decrease in loss metrics and a more meaningful correlation between features.

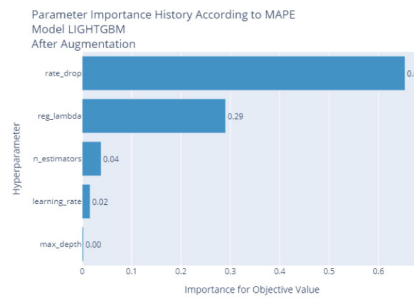


Table 5 Results of the optimization on the augmented dataset

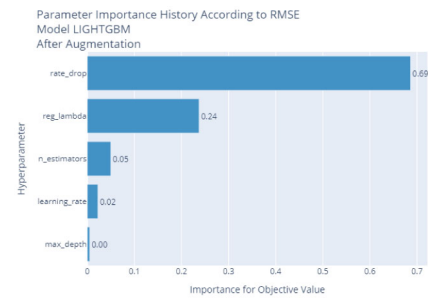
		Optimized for RMSE	Optimized for MAE	Optimized for MAPE
Learning rate	XGBoost	0.0333	0.0486	0.0444
	LightGBM	0.0457	0.0458	0.0452
Drop rate	XGBoost	0.3882	0.0014	0.8155
	LightGBM	0.0006	0.0005	0.0012
Number of estimator	XGBoost	125	224	248
	LightGBM	300	134	192
Req_Lambda	XGBoost	0.1	0.05	0.5
	LightGBM	0.55	0.65	1.05
Max depth	XGBoost	125	11	261
	LightGBM	259	243	120
Optimized loss	XGBoost	5.67	2.83	2.06
	LightGBM	5.82	3.35	2.48



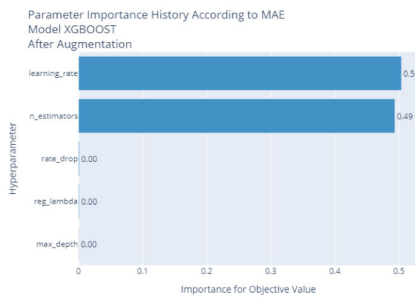
(a)



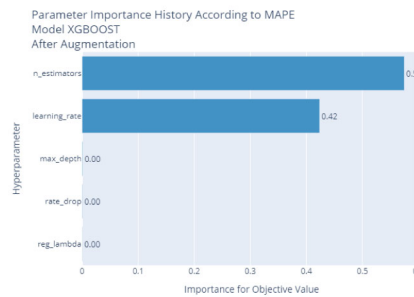
(b)



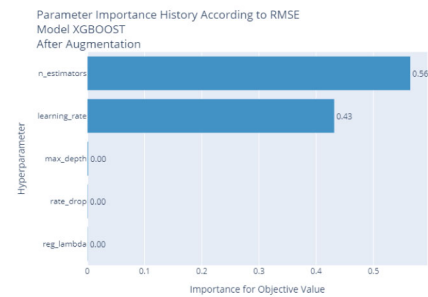
(c)



(d)



(e)



(f)

Fig. 12 Parameter importance of the models on the augmented dataset **a** LightGBM-MAE **b** LightGBM-MAPE **c** LightGBM-RMSE **d** XGBoost-MAE **e** XGBoost-MAPE **f** XGBoost-RMSE

- The effectiveness of optimizing the proposed ML model using the RMSE, MAE, and MAPE metrics was confirmed by the results that closely match the expected effects of the input parameters. In other words, the impact of the mix components representing the input data on the estimated compressive strength yielded results consistent with the data characteristics presented at the beginning of the study.

For example, the micro steel fibers' positive impact on the compressive strength's predicted value was accurately captured.

- Both the mean absolute SHAP values and the beeswarm plots visually showed the influence of all features on compressive strength before the augmentation. In particular, micro steel fibers and superplasticizers individually

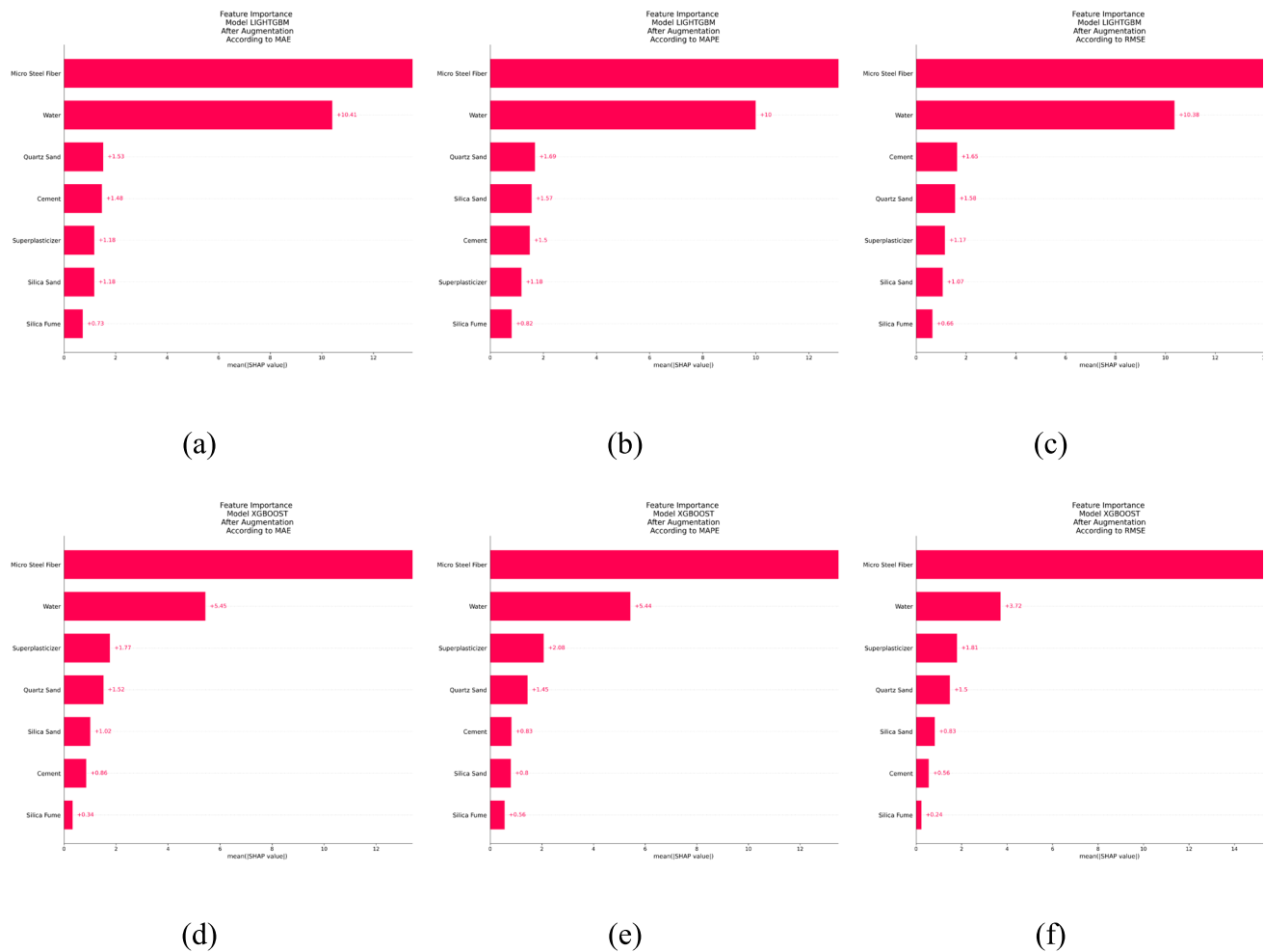


Fig. 13 Feature the importance of the models according to SHAP Values on the Augmented Dataset **a** LightGBM-MAE **b** LightGBM-MAPE **c** LightGBM-RMSE **d** XGBoost-MAE **e** XGBoost-MAPE **f** XGBoost-RMSE

showed the most pronounced and significant effects. Other features were more scattered. However, feature importances were more focused and consistent after augmentation, indicating more reliable and apparent impacts. The augmentation successfully reduced the variability and uncertainty in the ML models.

- In estimation problems, it is essential to determine the input parameters that affect the output parameters correctly. For example, when studying the effects of mix components on compressive strength, considering the impact of the individual components can be misleading. This study found that after augmentation, the effects of cement and silica fumes were relatively small compared to those of micro steel fibers, superplasticizers, and water. However, it is generally known that these components are critical for increasing compressive strength. Therefore, the relative proportions of some mix components should be considered when evaluating the estimation performance of the ML model for future work.

Consequently, this study integrates data augmentation, hyperparameter optimization, and SHAP-based model interpretation to enhance machine learning models' predictive accuracy and reliability in HPC applications. However, the small dataset limits the model's generalizability across different HPC mixtures, even when augmented. This suggests that exploring alternative augmentation techniques, such as



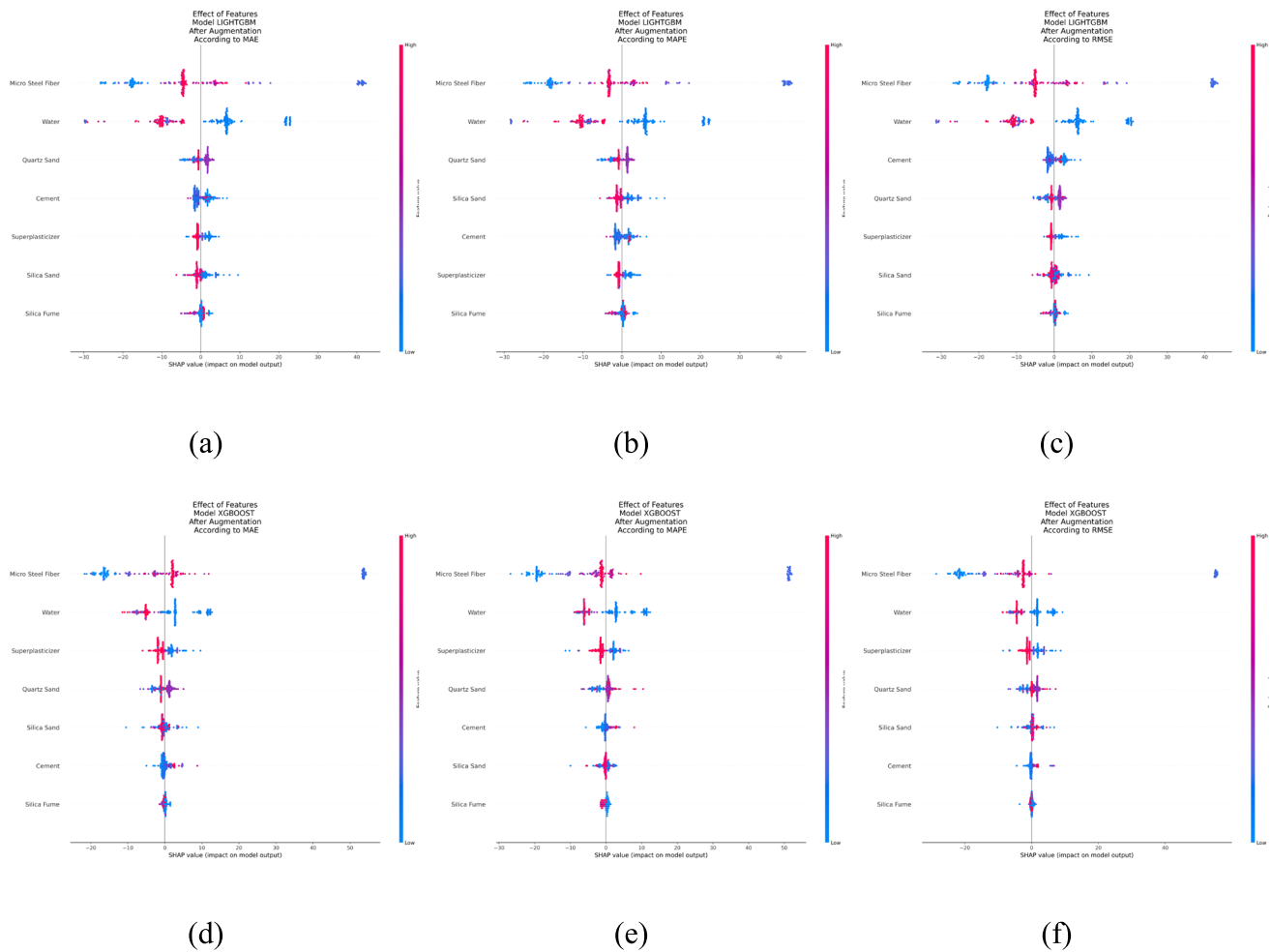


Fig. 14 Attributes effect on the output according to SHAP values on the augmented dataset **a** LightGBM-MAE **b** LightGBM-MAPE **c** LightGBM-RMSE **d** XGBoost-MAE **e** XGBoost-MAPE **f** XGBoost-RMSE

variational autoencoders (VAEs), may yield further improvements. Additionally, reliance on a specific set of input parameters could overlook other vital factors affecting compressive strength. Future research should focus on testing larger and more diverse datasets, exploring alternative augmentation methods, and applying these models to various types of concrete in real-world scenarios to enhance generalizability and practical relevance.

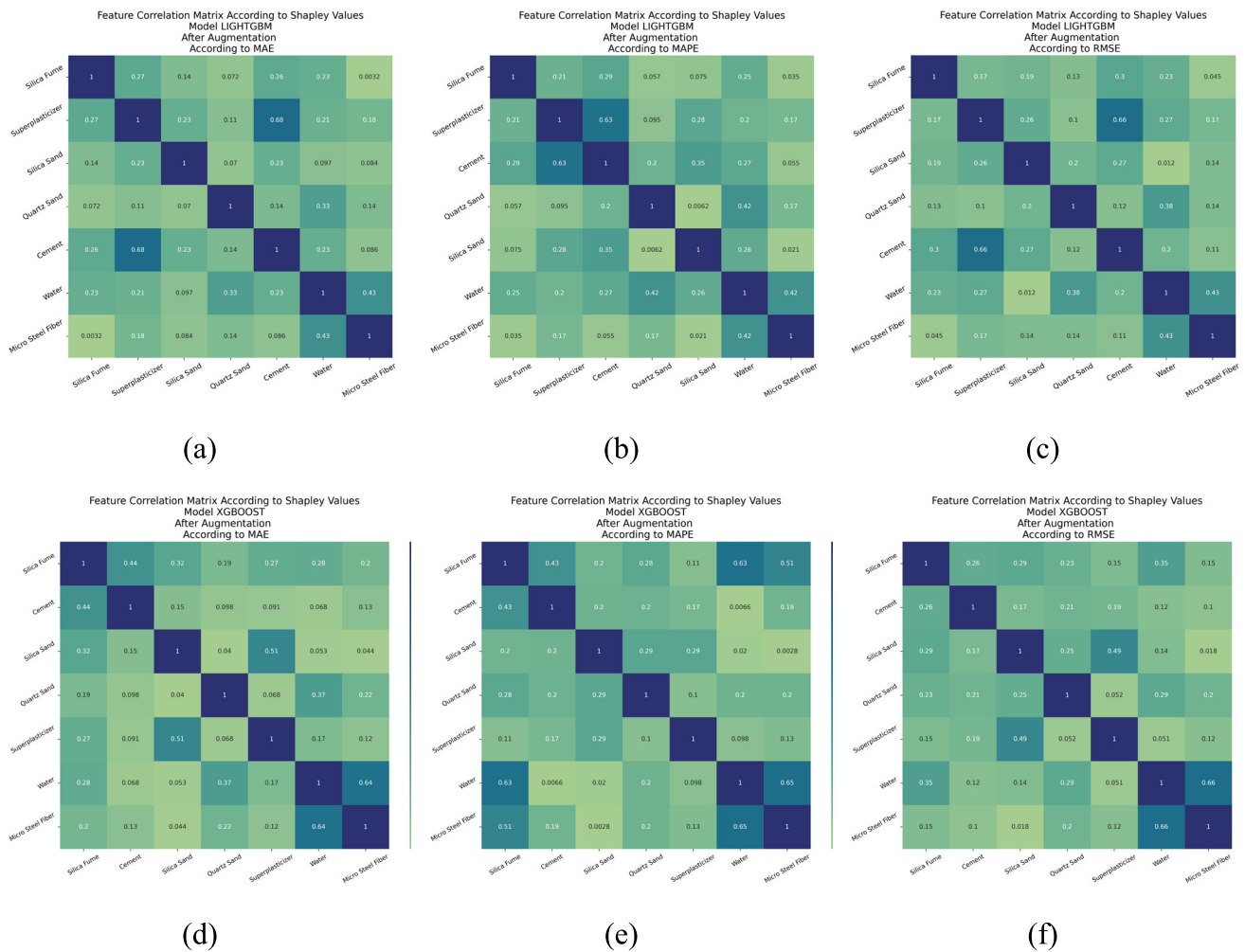


Fig. 15 Feature correlation matrix according to SHAP values on the augmented dataset **a** LightGBM-MAE **b** LightGBM-MAPE **c** LightGBM-RMSE **d** XGBoost-MAE **e** XGBoost-MAPE **f** XGBoost-RMSE

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Declarations

Conflict of interest The authors acknowledge that there are no known conflicts of interest related to this publication and that there was no significant financial support for this work that could have influenced its outcome.

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